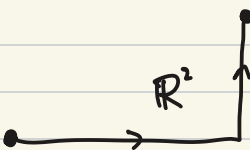


A comment on line integrals:

If $\vec{x}(t): I \rightarrow \mathbb{R}^n$ is a parametrized path and we want to measure its length, we need $\vec{x}(t)$ to be differentiable, and then we need to integrate $\int_I 1 \, ds = \int_I \|\vec{x}'(t)\| \, dt$.

But we don't need $\vec{x}(t)$ to be C^1 or differentiable on all I for this integral.

It suffices to take $\vec{x}: I \rightarrow \mathbb{R}^n$ continuous and piecewise differentiable

e.g.  $\vec{x}: [0, 2] \rightarrow \mathbb{R}^2$

$$\vec{x}(t) = \begin{cases} (t, 0) & 0 \leq t \leq 1 \\ (1, t) & 1 \leq t \leq 2 \end{cases}$$

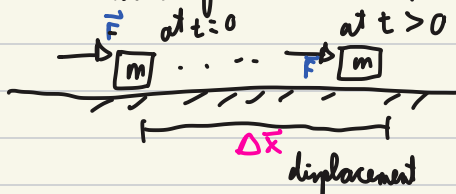
Vector Line Integrals

intuition: A scalar line integral $\int_{\vec{x}} f \, ds$ is
 "how much of the function f we are collecting while travelling the path $\vec{x}$ "

A vector line integral $\int_{\vec{x}} \vec{F} \cdot d\vec{s}$ is
 "how much of the vector field $\vec{F}$ is pushing us forward along our path as we travel the path $\vec{x}$ "

$$\int_{\vec{x}} \vec{F} \cdot d\vec{s} = \int_I \vec{F}(\vec{x}(t)) \cdot \vec{x}'(t) \, dt$$

e.g. Work - Constant force on a moving object



$$W = \vec{F} \cdot \Delta \vec{x}$$

if the force is changing or the path is not a straight line in the same direction of the force, then we need to integrate

$$W = \int_{\vec{x}} \vec{F} \cdot d\vec{s}$$

Suppose the mass m is moving horizontally *in the x-y plane* and the only forces are gravity $\vec{F}_g = -mg \hat{k}$ and normal force table pushing back up "against the force of gravity" $\vec{F}_n = mg \hat{k}$

Total work = 0 *forces are perpendicular to displacement*

Now consider friction $\vec{F}_\mu = \mu \|\vec{F}_n\| (-\vec{T})$ (directly proportional to normal force and with opposite direction to the instantaneous displacement)
 $\vec{T}(t) = \frac{\vec{x}'(t)}{\|\vec{x}'(t)\|}$ *unit vector in the direction of the displacement*

$$\int_{\vec{x}} (\sum \vec{F}) \cdot d\vec{s} = \int_{\vec{x}} (\vec{F}_g + \vec{F}_n + \vec{F}_\mu) \cdot d\vec{s}$$

$$= \int_{\vec{x}} \vec{F}_g \cdot d\vec{s} + \int_{\vec{x}} \vec{F}_n \cdot d\vec{s} + \int_{\vec{x}} \vec{F}_\mu \cdot d\vec{s}$$

$$= 0 + 0 + \int_I \mu \|\vec{F}_n\| \left(\frac{-\vec{x}'(t)}{\|\vec{x}'(t)\|} \right) \cdot \vec{x}'(t) dt$$

$$= -\mu \|\vec{F}_n\| \int_I \frac{\vec{x}'(t) \cdot \vec{x}'(t)}{\|\vec{x}'(t)\|} dt = -\mu \|\vec{F}_n\| \int_I \|\vec{x}'(t)\| dt$$

** $\vec{x}: I \rightarrow \mathbb{R}^3$*

$$= -\mu mg s \quad s = \text{length of the path } I$$

e.g. Compute the vector line integral of the field $\vec{F}(x,y,z) = (x, y^2, x+z)$ along the path $\vec{r}(t) = (2t+1, t, 3t-1)$ $0 \leq t \leq 1$.

$$\vec{v}(t) = \vec{r}'(t) = (2, 1, 3)$$

$$\vec{F}(\vec{r}(t)) = (2t+1, t^2, 2t+1+3t-1) = (2t+1, t^2, 5t)$$

$$\begin{aligned} \int_{\vec{r}} \vec{F} \cdot d\vec{s} &= \int_0^1 \vec{F}(\vec{r}(t)) \cdot \vec{v}(t) dt \\ &= \int_0^1 (2t+1, t^2, 5t) \cdot (2, 1, 3) dt \\ &= \int_0^1 (4t+2+t^2+15t) dt = \left. \frac{19}{2} t^2 + 2t + \frac{t^3}{3} \right|_{t=0}^{t=1} \\ &= \frac{19}{2} + 2 + \frac{1}{3} \end{aligned}$$

The differential form of a vector line integral

This is defined in all dimensions, but we will illustrate this for $\mathbb{R}^3$:

Let's say $\vec{r}(t) = (x(t), y(t), z(t))$ is a differentiable path and $\vec{F}(x,y,z) = M(x,y,z)\hat{i} + N(x,y,z)\hat{j} + P(x,y,z)\hat{k}$ is a vec. field.

$$\begin{aligned} \text{Then, } \int_{\vec{r}} \vec{F} \cdot d\vec{s} &= \int_a^b [M(x,y,z)x'(t) + N(x,y,z)y'(t) + P(x,y,z)z'(t)] dt \\ &= \int_{\vec{r}} M(x,y,z) dx + \int_{\vec{r}} N(x,y,z) dy + \int_{\vec{r}} P(x,y,z) dz \end{aligned}$$

differential form : $M(x,y,z)dx + N(x,y,z)dy + P(x,y,z)dz$

$$\text{e.g. } \vec{r}(t) = (t, 3, t^6) \quad t \in [0, 1] \quad \int_{\vec{r}} (x+y)dx + (y+z)dy + x^2dz$$

$$\begin{aligned} &\int_0^1 [(x+y)x'(t) + (y+z)y'(t) + x^2z'(t)] dt \\ &= \int_0^1 [(t+3)(1) + (3+t^6)(0) + t^2(6t^5)] dt \end{aligned}$$

Reparametrization (of a parametrized path)

Let's start by recalling the difference between

- Parametrized path:

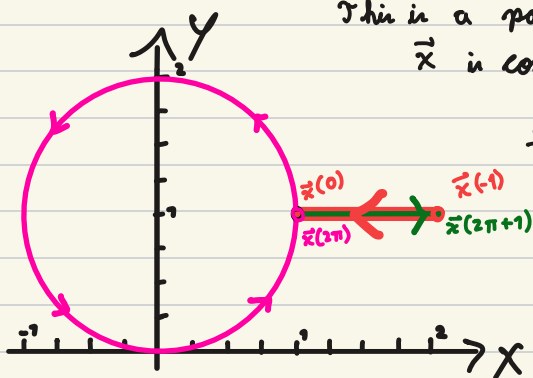
is a continuous function $\vec{x}: I \rightarrow \mathbb{R}^n$ (I is some interval, could be open, closed, ...)

* we also require $\vec{x}$ to be piecewise differentiable in order to compute arclength and other line integrals

e.g.
$$\vec{x}(t) = \begin{cases} (1-t, 1) & , -1 \leq t \leq 0 \\ (\cos t, 1 + \sin t) & , 0 < t \leq 2\pi \\ (1+t-2\pi, 1) & , 2\pi < t \leq 2\pi+1 \end{cases}$$

This is a parametrized path:

$\vec{x}$ is continuous and piecewise differentiable, but "the path is tracing back over its own steps".



* Parametrized paths are allowed to self-intersect and retrace their own steps

- Parametrization of a curve:



We start with a curve $C \subseteq \mathbb{R}^n$ and we parametrize C as $\vec{\gamma}: I \rightarrow \mathbb{R}^n$ cont. and piecewise diff. and only intersecting itself in finitely many points

def: Suppose $\vec{x}: [a, b] \rightarrow \mathbb{R}^n$ is a piecewise diff. path
and $\vec{y}: [c, d] \rightarrow \mathbb{R}^n$ piecewise diff. path
such that $\vec{x}$ and $\vec{y}$ trace the same curve in $\mathbb{R}^n$.

clp there is a function $f: [c, d] \rightarrow [a, b]$ that is
one-to-one, surjective
and
 $\vec{y}(t) = \vec{x}(f(t))$ for all $t \in [c, d]$
then we say $\vec{y}$ is a reparametrization of $\vec{x}$.

e.g. $\vec{x}(t) = (\cos t, \sin t) \quad 0 \leq t \leq \pi/2$

and $\vec{y}(t) = (\cos(2t), \sin(2t)) \quad 0 \leq t \leq \pi/4$

and $\vec{z}(t) = (\sin t, \cos t) \quad 0 \leq t \leq \pi/2$



Let's find $f: [0, \pi/4] \rightarrow [0, \pi/2]$
such that $\vec{y}(t) = \vec{x}(f(t))$
 $f(t) = 2t$

$$\vec{y}(t) = (\cos(2t), \sin(2t)) = \vec{x}(2t)$$

this shows $\vec{y}$ is a reparam. of $\vec{x}$

Let's find $g: [0, \pi/2] \rightarrow [0, \pi/2]$ such that $\vec{z}(t) = \vec{x}(g(t))$.
 $g(t) = \pi/2 - t$

$$\begin{aligned} \vec{z}(t) &= (\sin t, \cos t) = (-\sin(-t), \cos(-t)) \\ &= (-\cos(-t - \pi/2), \sin(\pi/2 - t)) \\ &= (\cos(\pi - t - \pi/2), \sin(\pi/2 - t)) \\ &= \vec{x}(g(t)) \end{aligned}$$

* $\sin t = -\sin(-t)$

$\sin(t + \pi/2) = \cos t$

$\sin t = \cos(t - \pi/2)$

$\cos(t + \pi) = -\cos t$